HARSHIT MUHAL

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EXPERIENCE

Microsoft **June 2025 – Present**
Software Engineer 2 *Hyderabad, India*

- Designed and built an **AI-powered API orchestrator** using **Semantic Kernel** that extended a RAG-based chatbot to handle transactional queries, enabling self-service access to operational data and reducing dependency on email-based support workflows.
- Developed automated incident remediation using **Azure SRE Agent** for high-frequency production issues, reducing resolution time by **90%** and saving **30+ on-call hours per month**.
- Integrated **streaming functionality** into chatbot APIs using the **Azure OpenAI GPT-4.0 model**, enabling real-time responses from **KB articles** through **prompt engineering**. Optimized system prompts, resolved performance bottlenecks, and reduced response time from **40s to 10s**, enhancing user experience and efficiency.

Microsoft **June 2022 – May 2025**
Software Engineer *Hyderabad, India*

- Engineered an Azure-based asynchronous processing solution for a Windows & XBOX service using **Azure Functions**, eliminating external API dependencies, reducing **$30K in annual Azure costs**, and scaling production load by **5x**. Optimized processing time from **7 minutes to 2 minutes** for **5K+ devices**.
- Diagnosed and resolved **30+ monthly API failures** in a **WCF microservice** by implementing **Azure Functions** for asynchronous processing, eliminating manual interventions and enhancing system resilience. Applied similar **cloud-based optimizations** across multiple microservices.
- Led the **end-to-end migration** of the **Cosmos DB SDK from V2 to V3**, modernizing authentication with **Managed Identities** and extending **secure access** across **Azure Service Bus, Event Hubs, and Storage Accounts**, improving security and maintainability.

Microsoft **10 May 2021 – 2 July 2021**
Software Engineer Intern *India*

- Developed a **rollback mechanism** for **Azure Cosmos DB documents**, enabling consuming services to swiftly **restore the data tier** in case of deployment failures, enhancing system **resilience and uptime**.

SKILLS & TECHNOLOGIES

Cloud & DevOps: Azure Active Directory, Microsoft Graph API, Azure Functions, Azure Key Vault, Azure Storage (Cosmos DB, Blob Containers, Storage Queues), Azure DevOps, Azure Service Bus, Azure Event Hubs
Programming: C#, .NET Framework, .NET Core, SQL **Databases:** Cosmos DB, SQL Server
AI: Azure OpenAI, Semantic Kernel, AI agents, prompt engineering, Retrieval-Augmented Generation

EDUCATION

Delhi Technological University B.Tech – Computer Engineering *Aug 2018 – May 2022* *CGPA: 9.53/10.0*

PUBLICATIONS

- Harshit Muhal, Gaurav Khatri, Gaurav Kumar Dhama, Deepika Bansal, "An Ensemble Approach for Credit Card Fraud Detection Using Champion-Challenger Analysis," IEEE International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES), 2022.
- Dr. Shailender Kumar, Harshit Muhal, Ishan Chaudhary, Kirti Dabas, "Identification of IUU Transshipment Activity Using AIS Data," IEEE 3rd International Conference for Emerging Technology (INCET), 2022.

PROJECTS

- **Image Captioning:** Developed a model that generates real-world image captions using ResNet (image encoding) and LSTM (text generation) in a supervised learning setup.
- **IMDb Review Sentiment Analysis:** Implemented multiple CNN-LSTM networks, achieving 89.39% accuracy on the IMDb dataset. One used combined kernels from multiple branches, while another utilized residual connections.

ACHIEVEMENTS & AWARDS

EC Excellence Ace Award: Recognized for **top performance** and consistently exceeding expectations at Microsoft.

EC Learn-It-All Award: Honored for continuous learning and **technical competency enhancement** at Microsoft.

Google Hash Code 2021: Global Rank **1442**, Country Rank **244**.

Google Kickstart: Round B 2021 (Global Rank **609**), Round G 2020 (Global Rank **844**)(Username – harshitmuhal).

Codeforces: Round 720 (Global Rank **641**), Round 719 (Global Rank **478**)(Username – Infinite.Loop).

Achieved **Top 7 rank** in department out of **450 students** in B.Tech at DTU.